

What is BMR(Basal Metabolic Rate)

Basal Metabolic Rate (BMR) is the minimum amount of energy (measured in kcal/day or kJ/day) that your body requires to maintain vital physiological functions at complete rest, such as:

Breathing

Blood circulation

Cell production

Maintenance of body temperature

Brain & nerve functioning

Hormonal activities

It represents the energy expenditure necessary for survival, without any physical activity.

CONDITIONS FOR BMR MEASUREMENT

To get an accurate BMR measurement, it is taken under:

- Post-absorptive state (usually 12–14 hours after last meal)
- Thermoneutral environment (no shivering or sweating)
- Complete physical & mental rest
- After a full night's sleep
- In a reclining position
- These conditions ensure the body is doing only the minimal work needed to stay alive.

CALCULATION OF BMR

To calculate Basal Metabolic Rate (BMR), several equations are used in physiology and nutrition science. The two most commonly used standardized equations are:

Harris–Benedict Equation

The Harris–Benedict Equation is a classical predictive formula developed in 1918–1919 to estimate an individual's Basal Metabolic Rate (BMR) based on weight, height, age, and gender. It was one of the earliest scientific methods for quantifying resting energy expenditure and became widely used in nutrition, physiology, and clinical settings for determining daily caloric needs.

For Men:

$$BMR = 13.397W + 4.799H - 5.677A + 88.362$$

For Women:

$$BMR = 9.247W + 3.098H - 4.330A + 447.593$$

- Where:
- W = weight in kilograms (kg)
- H = height in centimeters (cm)
- A = age in years

CALCULATION OF BMR

Advantages of Harris–Benedict Equation

- Simple and non-invasive method for estimating BMR.
- Requires only basic anthropometric data (age, weight, height, gender).
- Historically important and widely referenced in nutrition and physiology research.
- Useful for diet planning and estimating total daily energy expenditure (TDEE).
- Good for population-level or classroom estimations where accuracy requirements are moderate.
- Helps understand the relationship between body size, age, and metabolism.

CALCULATION OF BMR

Limitations of Harris–Benedict Equation

- Developed in early 20th century, based on populations with different body composition profiles.
- Tends to overestimate BMR in modern populations.
- Does not account for lean body mass or body composition, which strongly influences metabolism.
- Accuracy decreases in:
 - obese individuals
 - athletes/muscular individuals
 - elderly
 - clinical/metabolic disorders
- Less preferred in clinical dietetics, where Mifflin–St Jeor provides better predictive validity.

CALCULATION OF BMR

Mifflin–St Jeor Equation (More Accurate, Modern Standard)

The Mifflin–St Jeor Equation, developed in 1990, is a modern predictive formula used to estimate Basal Metabolic Rate (BMR) based on age, weight, height, and gender. It was designed to reflect changes in body composition, lifestyle, and metabolic patterns of contemporary populations. Due to its higher accuracy and lower bias, it is widely used today in clinical nutrition, obesity management, and sports diet planning.

For Men:

$$BMR = 10W + 6.25H - 5A + 5$$

For Women:

$$BMR = 10W + 6.25H - 5A - 161$$

CALCULATION OF BMR

Advantages of Mifflin–St Jeor Equation

- More accurate than Harris–Benedict, especially for modern populations.
- Validated for both non-obese and obese individuals.
- Suitable for clinical dietetics and nutrition planning.
- Lower tendency to overestimate BMR.
- Simple and requires only basic anthropometric measurements.
- Recommended by many dietitians, nutritionists, and sports physiologists.

CALCULATION OF BMR

Limitations / Disadvantages of Mifflin–St Jeor Equation

- Still an estimation, not a direct measurement.
- Like other predictive equations, it doesn't account for:
 - lean body mass
 - muscle mass
 - hormonal changes
 - metabolic diseases
- Can underestimate BMR in highly muscular individuals (e.g., athletes).

CALCULATION OF BMR

- Eg. Calculation using Mifflin–St Jeor Equation

Let's assume a female:

- Weight = 60 kg
- Height = 165 cm
- Age = 25 years

Using **Mifflin–St Jeor**:

$$\begin{aligned} BMR &= 10(60) + 6.25(165) - 5(25) - 161 \\ &= 600 + 1031.25 - 125 - 161 \\ &= 1345.25 \text{ kcal/day} \approx 1345 \text{ kcal/day} \end{aligned}$$

CALCULATION OF BMR

Simple Approx Estimate

$$BMR \approx 24 \times \text{Body Weight (kg)}$$

(This assumes avg BMR $\approx$ 1 kcal/kg/hour)

E.g., 60 kg $\times$ 24 = 1440 kcal/day

BMR VALUES

Normal BMR Values

Adult males: ~1600–1800 kcal/day

Adult females: ~1200–1500 kcal/day